

Agenda

1. Announcements

- Exam 1 is tomorrow!
- No HW due on Wednesday (four assignments due next week)
- No discussion quiz on Thursday
- My office hours are in Van Vleck B215 today, 1:00-2:30

2. Exam Logistics

3. Section 9.4

Exam Logistics

Midterm 1, Tuesday February 28, 7:30pm - 9:00 pm

Midterm 1: Tuesday February 28, 7:30-9:00 pm

Discussion Section	TA Name(s)	Exam Room Assignment
321, 323	Anuk	Van Vleck B239
322, 324, 325, 326, 327	Logan, Jiaming, and Jiaqi	Van Vleck B102
328, 329, 330, 332	Hongjing and Kevin	Van Vleck B130

Exam Info Page:

go.wisc.edu/1xxzfx

1. Please try to arrive at least 5 minutes early.
2. Please sit down so that there is exactly one empty seat between you and the person next to you (otherwise we may ask you to move)
3. You may leave (quietly) when you finish the exam. However, we'll ask everyone still in the room with 5 minutes remaining to stay until the end. This way, everyone can finish in relative quiet.
4. You will be required to include the following on your exam cover sheet printed legibly:
 - Name (as it appears on Canvas)
 - Wisc.edu email address
 - My last name (Lindsey)
 - Your TA name

Section 9.4 - Methods for Population Growth

Let P be the population of an organism.

Let t be time.

Natural Growth - "populations that grow proportional to size."

$$\frac{dP}{dt} = kP, \text{ where } k \text{ is a constant and } P_0 = P(0) \text{ is the initial population size.}$$

The constant k is the relative growth rate.

From 9.1, the solutions:

$$P(t) = P_0 e^{kt} \quad (k = \text{constant}, P_0 = P(0))$$

Activity: E. Coli has a known relative growth rate

$K = 1.06$ bacteria/hour. Suppose a bacteria colony starts with 1200 bacteria. Find a model for the size of the colony.
 $P_0 = 1200$

$$P = \underbrace{1200}_{P_0} e^{\frac{1.06(1)}{K} \frac{t}{t}}$$

Logistic Model

A more complex model is the logistic model given by the differential equation $\frac{dP}{dt} = kP(1 - \frac{P}{M})$

M is a constant, the carrying capacity (some limiting factor) where the population levels out.

$$\Rightarrow P(t) = \frac{M}{1 + A e^{-kt}}$$

$A = \frac{M - P_0}{P_0}$

Activity Let P be the size of a fish species' population

Let t be time (in years). Suppose the population is modeled by $\frac{dP}{dt} = 0.40 P(1 - \frac{P}{1200})$ with an initial population of 50 fish.

- Write a model for the population $P(t)$.
- What will the population be after 10 years?
- When will the population reach 1000 fish?

$$a) A = \frac{1200 - 50}{50} = 23$$

$$P(t) = \frac{1200}{1 + 23 e^{-0.4t}}$$

$$b) \quad p(10) = \frac{1200}{1 + 23e^{(-4)}} \approx 844 \text{ fish}$$

$$c) \quad 1000 = \frac{1200}{1 + 23e^{-.4t}}$$

$$1000(1 + 23e^{-.4t}) = 1200$$

$$1 + 23e^{-.4t} = 1.2$$

$$23e^{-.4t} = 0.2$$

$$e^{-.4t} = \frac{1}{115}$$

$$\ln(e^{-.4t}) = \ln\left(\frac{1}{115}\right)$$

$$-.4t = \ln 1 - \ln 115$$

$$-.4t = -\ln 115$$

$$t = \frac{\ln(115)}{0.4} \approx \boxed{11.862 \text{ years}}$$

$$\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$$

M is a constant, the carrying capacity
(some limiting factor)
where the population
levels out.

$$\Rightarrow P(t) = \frac{M}{1 + \underbrace{A}_{\frac{M-P_0}{P_0}} e^{-kt}}$$

$$\frac{dP}{dt} = 80K \left(1 - \frac{80}{2000}\right)$$

$$\int dP = \int 80K \left(1 - \frac{80}{2000}\right) dt$$

$$80K - \frac{640K}{2000}$$

$$P = t \left(80K - \frac{640K}{2000}\right) + C$$